

Cybersecurity & Digital Safety

> Launching the Next Gen of Tech
Leaders

A comprehensive middle-school elective proposal
focused on ethical hacking, coding, and professional
portfolio building.



The Opportunity

Why start in Middle School?

| EDUCATIONAL GOALS



Digital Safety

Mastering responsible digital citizenship, phishing defense, and privacy management.



Tech Literacy

Navigating Linux systems, networking protocols, and Python automation.



Problem Solving

Building logic and persistence through gamified cybersecurity puzzles.

THE "WHITE HAT" MISSION

This course emphasizes ethical hacking—using tech skills for good. We transition students from mindless screen time to active, secure development.

By ages 12–15, students are ready to understand the legal boundaries and professional standards of the tech industry.

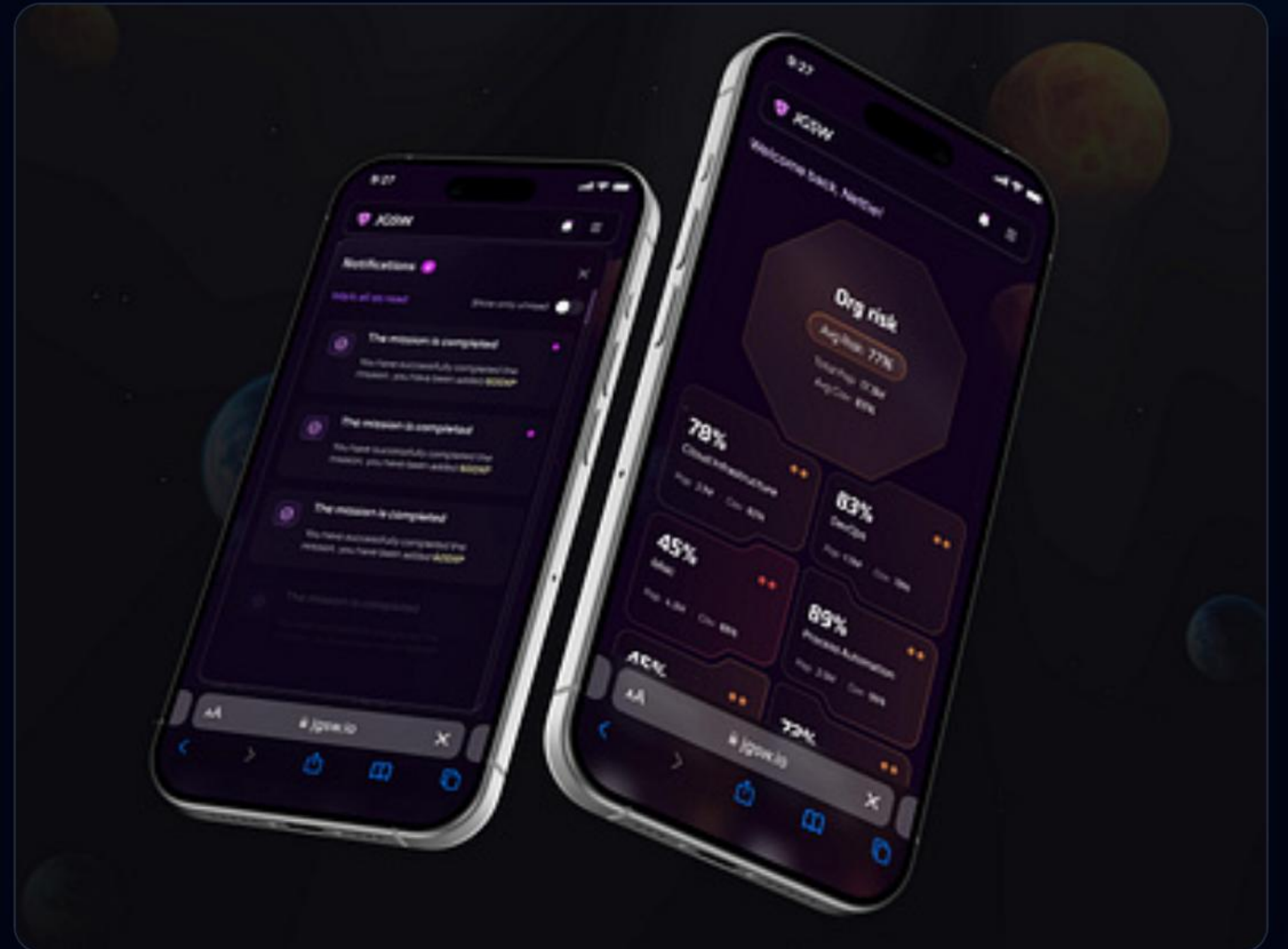


16-WEEK COURSE ROADMAP

PHASE	WEEKS	CORE FOCUS
Digital Safety	1–4	Internet Safety, Networking & Phishing Awareness
The Power of Linux	5–8	Terminal Navigation & Cybersecurity Defense
Automation & Web	9–12	Python Programming & Web Security Basics
Portfolio & CTF	13–16	Capture The Flag Labs & GitHub Showcase

HANDS-ON LEARNING TOOLS

- **TryHackMe:** Gamified labs that make learning feel like a video game.
- **OverTheWire:** Text-based wargames for mastering the Linux command line.
- **Industry Standards:** Learning how the pros actually work in terminal environments.



The GitHub Portfolio

Building a Digital Legacy

| PROFESSIONAL DOCUMENTATION

Verifiable History

Students document every project with screenshots and technical notes. This creates a time-stamped history of their learning journey.

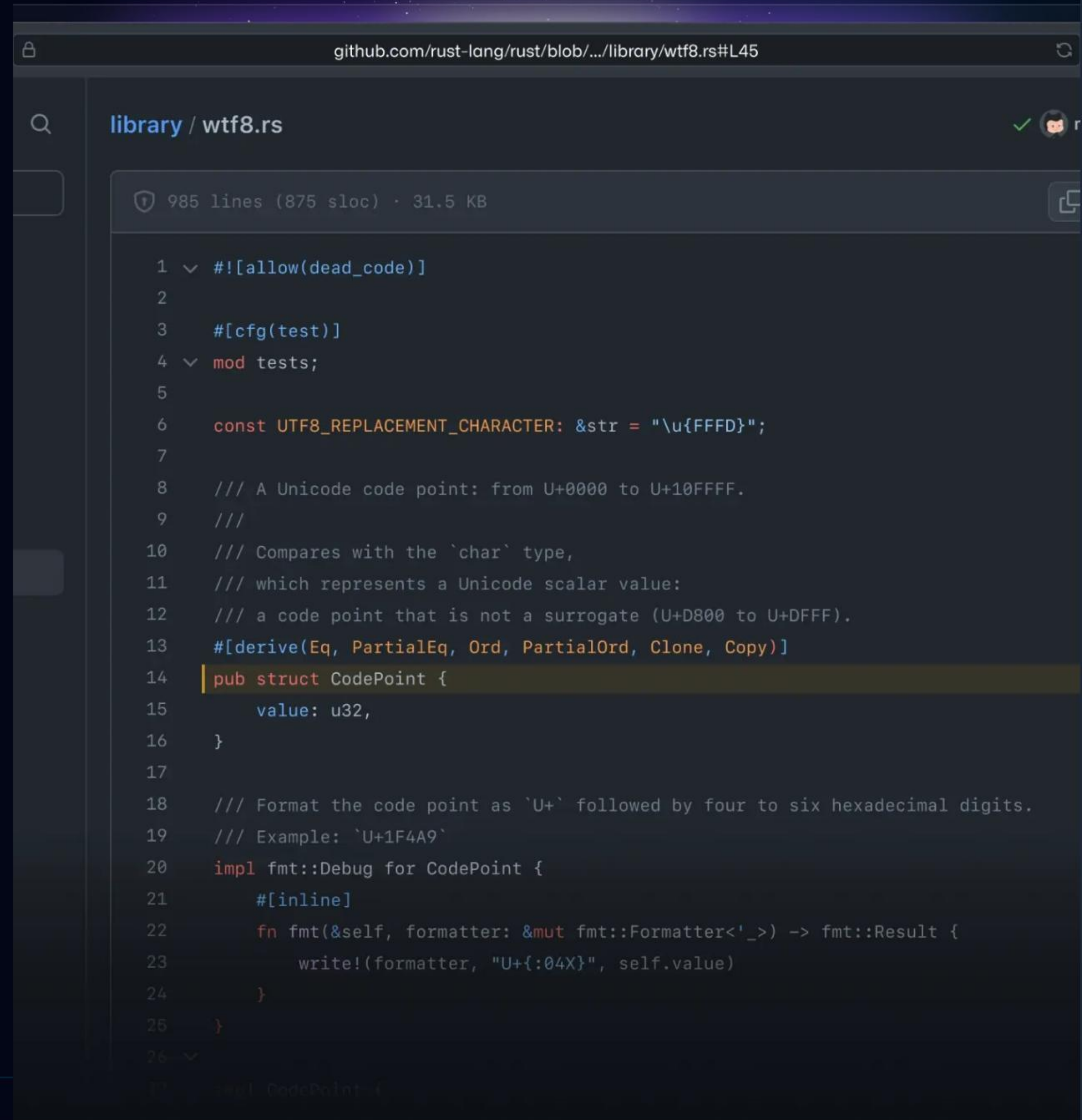
Open Source Ethics

By using the **MIT License**, students learn about intellectual property and how to participate in global tech communities.

THE LIVING RESUME

A structured GitHub profile is more powerful than a diploma for tech internships.

Students graduate this semester with a professional-grade repository ready for future college apps or jobs.



The screenshot shows a GitHub web interface for the repository 'library / wtf8.rs'. The browser address bar displays 'github.com/rust-lang/rust/blob/.../library/wtf8.rs#L45'. The repository page header indicates '985 lines (875 sloc) · 31.5 KB'. The code is displayed in a dark-themed editor with line numbers on the left. The code includes Rust syntax for configuration, tests, constants, and a struct definition. The line containing the struct definition is highlighted.

```
1  ▾ #[allow(dead_code)]
2
3  #[cfg(test)]
4  ▾ mod tests;
5
6  const UTF8_REPLACEMENT_CHARACTER: &str = "\u{FFFD}";
7
8  /// A Unicode code point: from U+0000 to U+10FFFF.
9  ///
10 /// Compares with the `char` type,
11 /// which represents a Unicode scalar value:
12 /// a code point that is not a surrogate (U+D800 to U+DFFF).
13 #[derive(Eq, PartialEq, Ord, PartialOrd, Clone, Copy)]
14 pub struct CodePoint {
15     value: u32,
16 }
17
18 /// Format the code point as `U+` followed by four to six hexadecimal digits.
19 /// Example: `U+1F4A9`
20 impl fmt::Debug for CodePoint {
21     #[inline]
22     fn fmt(&self, formatter: &mut fmt::Formatter<'_>) -> fmt::Result {
23         write!(formatter, "U+{:04X}", self.value)
24     }
25 }
26 ▾
27 impl CodePoint {
```


| THE CAREER ADVANTAGE

5+

YEARS HEAD START

Early Professionalization

Students entering high school with an established GitHub history and Linux experience are in the top 1% of their peer group. This course sets them on the path for high-paying STEM careers before they even graduate.

| FINAL STUDENT OUTCOMES

- **Technical Mastery:** Beginner-level Linux & Python command line knowledge.
 - **Digital Sovereignty:** Advanced understanding of encryption and personal safety.
 - **Professional Output:** A polished GitHub portfolio with documented lab completions.
 - **Ethical Foundation:** Clear understanding of legal and ethical technology practices.
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Questions?

Let's Secure the Future Together

`github.com/your-curriculum-repo`

Preparing students for a safe & successful tech career.

IMAGE SOURCES



https://img.magnific.com/free-vector/abstract-secure-technology-background_23-2148357087.jpg

Source: www.magnific.com



[https://s3.amazonaws.com/m.definedlearning.com/marketing-site/images/blog/Untitled%20design%20\(6\)-1.png](https://s3.amazonaws.com/m.definedlearning.com/marketing-site/images/blog/Untitled%20design%20(6)-1.png)

Source: blog.definedlearning.com



<https://cdn.dribbble.com/userupload/43657288/file/original-d182c3e64ec4c8f839cddc52c30cf71d.jpg?resize=400x300>

Source: dribbble.com



<https://images.ctfassets.net/8aevphvgewt8/4zdWojhLRcNgdkRQn7IBG4/3467a5833e1b04ac12fd7f608d26dadab/code-search-hero.webp?fm=webp>

Source: github.com
